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C2D2: Clean-to-Degraded Two-Pass Distillation for Robust Multi-Sensor Semantic Segmentation with a Frozen Vision Foundation Model

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ABSTRACT Multimodal semantic segmentation combines a camera with depth, event, and laser range sensors so that driving scenes are understood across weather, lighting, and sensor failures. The most accurate fusion models, however, lose much of their accuracy when a sensor fails at test time. We propose Clean-to-Degraded Two-Pass Distillation (C2D2), a training scheme that uses every batch twice: once with clean inputs and once with sensor-specific synthetic degradation, where the degraded pass is distilled from a frozen teacher trained on clean inputs. C2D2 needs no condition or degradation labels and adds no inference cost. We train with it a fusion model built on a frozen vision foundation backbone shared by all sensors, with sensor-specific low-rank adapters, cross-sensor attention, and a high-resolution decoding path. On the DELIVER benchmark, the model reaches 57.7 mean intersection over union on the test set, averaged over three training runs, one point above the best published four-sensor fusion model evaluated under the same protocol. When sensors are removed, its score averaged over all sensor subsets is 48.7, compared with 31.1 for the official weights of that model and at most 46.5 for strong fusion models retrained with the same degradation. We further analyse sensor dependence across conditions, thin objects, and failure cases [TODO: one sentence on MUSES and MCubeS once measured (FU-08)].

INDEX TERMS Knowledge distillation, multimodal semantic segmentation, robustness, sensor fusion

I. INTRODUCTION

AUTONOMOUS vehicles and mobile robots carry several sensors because each one fails in different conditions. A camera loses contrast at night and in fog, a depth estimate develops holes on reflective surfaces, an event camera goes silent on static scenes, and LiDAR returns thin out in rain and with distance. Multimodal semantic segmentation fuses these sensors so that the scene is still understood when one of them degrades.

The fusion models that lead the full-sensor benchmarks [1]–[6] handle adverse conditions by telling the fusion what the condition is. CAFuser [5] infers a condition token from the camera image and supervises it with a textual description of the weather and lighting, and DGFusion [6] adds local depth tokens and depth supervision. These models see the adverse scenes contained in their training data, but they are not trained with degraded or missing sensors, and they lose much of their accuracy when a sensor actually fails: under one evaluation protocol on DELIVER, the official DGFusion weights fall from 56.7 to 31.1 mIoU when averaged over all subsets of removed sensors. Methods designed for missing sensors [7]–[10] train with modality dropout and distillation, but the two lines of work rarely evaluate the same models under the same protocol [11]. Frozen vision foundation models offer strong features for every sensor, yet their multimodal adaptations mostly keep RGB as the primary stream and inject one auxiliary sensor [12].

We ask whether a change of the training signal, instead of additional condition supervision, can give an accurate fusion model the ability to hold up when its sensors degrade. We propose Clean-to-Degraded Two-Pass Distillation (C2D2). Every training batch is used twice, once with clean inputs and once with sensor-specific synthetic degradation, such as blur and exposure changes for the camera, holes for depth, dropped beams for LiDAR, and lost resolution for events, together with entire sensor removal. The degraded pass is supervised by the labels and distilled from a frozen teacher that sees the clean input. C2D2 needs no condition or degradation labels and adds no computation at inference. We apply it to a fusion model that reads every sensor with one

frozen DINOv3 backbone [13] through sensor-specific low-rank adapters, fuses the sensors with cross-modal attention, and recovers thin structures with a high-resolution decoding path.

On DELIVER [1], the model reaches 57.74 mIoU on the test set (mean of three training runs), 1.0 point above the official weights of DGFusion, the best published four-sensor fusion model, evaluated under the same protocol, and it keeps 48.7 mIoU when averaged over all subsets of removed sensors, compared with 31.1 for the official DGFusion weights and at most 46.5 for strong fusion models retrained with the same degradation [TODO: MUSES and MCubeS results (FU-08)]. We also analyse how the dependence on each sensor changes across conditions, where the high-resolution path helps, and where the model still fails.

Our contributions are as follows.

- We propose C2D2, a label-free two-pass training scheme that combines sensor-specific synthetic degradation with distillation from a clean-input teacher and adds no inference cost. [TODO: after FU-03 (L7) and FU-04 (DGFusion/CAFuser transfer): one sentence on how it differs from missing-modality distillation and whether its gain depends on the frozen foundation backbone]
- We build a single fusion model on a frozen vision foundation backbone that, trained with C2D2, improves full-sensor accuracy on DELIVER [TODO: and MUSES and MCubeS (FU-08)] while remaining robust under the public missing-sensor protocol.
- We analyse sensor dependence across conditions, missing, partial, and noisy sensors, thin objects, and failure cases, and report where the model still fails. [TODO: state the code and checkpoint release]

II. RELATED WORK

A. MULTIMODAL FUSION FOR SEGMENTATION

Early RGB-X segmentation fused one auxiliary sensor into an RGB stream, for example by cross-modal feature rectification [2] or by exchanging tokens between modality-specific transformers [14]. CMNeXt [1] generalized this to an arbitrary number of sensors with a hub that aggregates the auxiliary sensors before injecting them into RGB, and introduced DELIVER. Later work improved efficiency and accuracy with adapter-based early fusion [3] and pixel-level token exchange [4]. CAFuser [5] conditions the fusion on the scene condition inferred from the camera image, and DGFusion [6] uses depth, with depth supervision, to guide the fusion; both report leading results on DELIVER and MUSES. These methods are designed and evaluated with all sensors present. We re-train two of them with degraded inputs and show that their robustness depends mostly on training, not on their fusion design.

B. ROBUSTNESS TO MISSING AND DEGRADED SENSORS

A second line of work trains models that tolerate missing sensors. MAGIC [7], [15] treats all modalities equally and, during training, ranks modality features by their similarity to an aggregated feature. AnySeg [8] names the problem unimodal bias, the over-reliance on a dominant, faster-to-learn modality, and reduces it by distilling a multimodal teacher into a student trained with random modality dropout; functional-entropy regularization [16] targets the same bias. RMMSS [9] combines prototype distillation with feature selection. A recent benchmark [11] standardized entire-missing, random-missing, and noisy-modality evaluations, which we adopt, and attributed robustness differences to fusion design. Our study differs from these works in its question and its controls: rather than proposing another robust architecture, we take the accuracy-leading fusion models, hold their architecture fixed, change only the training data, and measure how their dependence on each sensor changes across conditions. FuseAdapt [10] adapts a frozen backbone with parameter-efficient fusion modules for missing modalities. Our training scheme uses the same building blocks as several of these methods, namely synthetic degradation and distillation from a clean teacher. We do not claim it as a new method; we use it to show where over-reliance comes from and to obtain a model that is both accurate and robust. Methods in this line usually skip a missing input, whereas the benchmark protocol replaces it with zeros; we keep the two families apart when comparing numbers.

C. ADAPTING FROZEN FOUNDATION MODELS

Vision foundation models are increasingly used as frozen encoders for dense prediction. The Segment Anything Model [17] has been adapted to multimodal inputs with adapters that inject auxiliary-sensor features into the RGB stream [12] or with mixtures of LoRA experts [18], and MemorySAM [19] reuses the memory mechanism of SAM 2 to fuse modalities. DINOv3 [13] provides dense features and a segmentation adapter derived from ViT-Adapter [20], which we reuse for our high-resolution path. Our backbone, LoRA adaptation [21], and query head [22] follow this practice; we treat them as the platform for our study, not as contributions.

III. SENSORS, BENCHMARKS, AND PROTOCOL

A. SENSORS AND THEIR FAILURE MODES

A driving platform rarely loses all sensors at once; it loses one, partially or completely, under conditions that the other sensors tolerate. An RGB camera gives dense appearance and texture but degrades at night, under low or excessive exposure, and in fog. A depth sensor, or depth reconstructed from stereo or LiDAR, gives geometry that is unaffected by illumination but contains holes on reflective, transparent, and distant surfaces. An event camera reports per-pixel brightness changes with high temporal resolution and dynamic range, but it is silent on static scenes and noisy in low light. A LiDAR gives accurate range on a sparse set of beams that shrinks with distance and is thinned by rain, fog, and absorbing surfaces. Radar, available in one of our benchmarks, measures range and velocity through adverse weather at a very low angular resolution. Each failure is therefore local to one sensor and correlated with a scene condition, which is the situation that

TABLE 1. Benchmarks used in this study.

Benchmark	Sensors	Classes	Train / val / test
DELIVER [1]	RGB, depth, event, LiDAR	25	3983 / 2005 / 1897
MUSES [23]	RGB, event, LiDAR, radar	19	1500 / 250 / 750
MCubeS [24]	RGB, AoLP, DoLP, NIR	20	302 / 96 / 102

MUSES test labels are withheld; test scores come from the official evaluation server.

a fusion model has to handle without being told which sensor has failed.

B. BENCHMARKS

We evaluate on three public benchmarks that cover synthetic and real sensor suites (Table 1). DELIVER [1] is a synthetic driving benchmark with RGB, depth, event, and LiDAR inputs and 25 classes. Its scenes span five weather and lighting conditions (sun, cloud, rain, fog, and night) and five sensor-failure cases (motion blur, over- and under-exposure, LiDAR jitter, and low-resolution events). Because every scene is rendered, the depth input is complete and noise-free, which turns out to matter for how fusion models learn (Section VI). MUSES [23] is a real driving dataset recorded under clear, foggy, rainy, and snowy conditions by day and night, with an RGB frame camera, an event camera, a MEMS LiDAR, and an FMCW radar, all projected into the camera view, and the 19 Cityscapes classes. We use all four sensors. MCubeS [24] is a real material-segmentation dataset with RGB, angle and degree of linear polarization (AoLP, DoLP), and near-infrared (NIR) images and 20 material classes. It tests whether the same model transfers to non-driving sensors whose failure modes differ from those above.

C. EVALUATION PROTOCOL

Clean accuracy. We report mean intersection over union (mIoU). For each training run we keep the checkpoint with the best validation score and report its test score, without selecting checkpoints on the test set. On DELIVER we evaluate at 1024×1024 with batch size one against the native-resolution labels, and resample labels with nearest-exact interpolation. This detail matters: floor-aligned nearest resampling shifts predictions by half a pixel and lowered DELIVER test mIoU by 1.2 to 1.3 points in our measurements, with larger losses on thin classes, whereas center-aligned resampling reproduces the scores of the baselines' own conversion code. We therefore use center-aligned resampling for every model. Where a baseline reports a different protocol, we list both its published number and our re-evaluation (Table 2). MUSES is scored by its official evaluation, and MCubeS on the commonly used 102-image test split. We report the mean over three training seeds together with the best seed, because single runs on these benchmarks vary by up to one point. [TODO: MCubeS checkpoint selection: our loader evaluates the 102 test images during training (semseg/datasets/mcubes.py:142, val -> test.txt), as in common practice for this benchmark, so the sentence above about not selecting checkpoints on the test set does not hold for MCubeS. Decide (method): select on the 96 validation images, or report final-epoch checkpoints and state the practice.]

Missing and degraded sensors. We follow the definitions of a recent robustness benchmark for multimodal segmentation [11]. In the entire-missing-modality (EMM) setting, one or more sensors are replaced by zeros after input normalization, and the score is averaged over all 15 non-empty subsets of the four sensors. In the random-missing-modality (RMM) setting, the sensors that a subset would remove are instead corrupted by zeroing a random fraction $r \in \{0.25, 0.5, 0.75\}$ of their values, and the score is again averaged over the 15 subsets. In the noisy-modality (NM) setting, salt-and-pepper noise of density $D \in \{0.05, 0.1, 0.2\}$ corrupts every sensor at the same pixel positions. The benchmark additionally adds Gaussian noise in this setting; we report salt-and-pepper alone so that the corruption is identical for all sensors [TODO: confirm that every NM number in Table 5 was produced with the salt-and-pepper-only mode (tools/missing_modality_eval.py, default nm mode)]. Methods that skip a missing input instead of zero-filling it [8]–[10] answer a different question, so we do not mix the two families in one comparison. Unless stated otherwise, every robustness number for our model and for the re-trained baselines in this paper comes from the same evaluation code, the same images, and the same random seed.

IV. METHOD

We propose Clean-to-Degraded Two-Pass Distillation (C2D2), a training scheme for multi-sensor segmentation, and apply it to a fusion model built on a frozen vision foundation backbone (Fig. 1). We first describe the model (Sections IV-A–IV-E) and then C2D2 (Section IV-F).

A. FROZEN BACKBONE WITH SENSOR-SPECIFIC ADAPTERS

Let x_s denote the input of sensor $s \in \{1, \ldots, S\}$, converted to a three-channel image (Section V). Every x_s is processed by the same DINOv3 ViT-L/16 backbone [13], whose weights are frozen. Each sensor has its own low-rank adapters (LoRA) [21] of rank 8 on the query and value projections of every attention block, so that each sensor can adapt the shared features to its own statistics. We take the token maps after blocks 6, 12, 18, and 24 and project each with a sensor-specific 1×1 layer, which gives a stride-16 token map T_s per sensor.

B. CROSS-MODAL FUSION

Each sensor attends to all other sensors. In each of two layers, the tokens of sensor s form the queries of a multi-head attention whose keys and values are the tokens of the other sensors,

$$\hat{T}_s \leftarrow \hat{T}_s + \mathrm{MHA}\big(\hat{T}_s,\ [\hat{T}_{s'}]_{s' \neq s}\big), \qquad \hat{T}_s^{(0)} = T_s, \quad (1)$$

followed by a feed-forward layer. The attention weights are shared across sensors, so every sensor is treated by the same

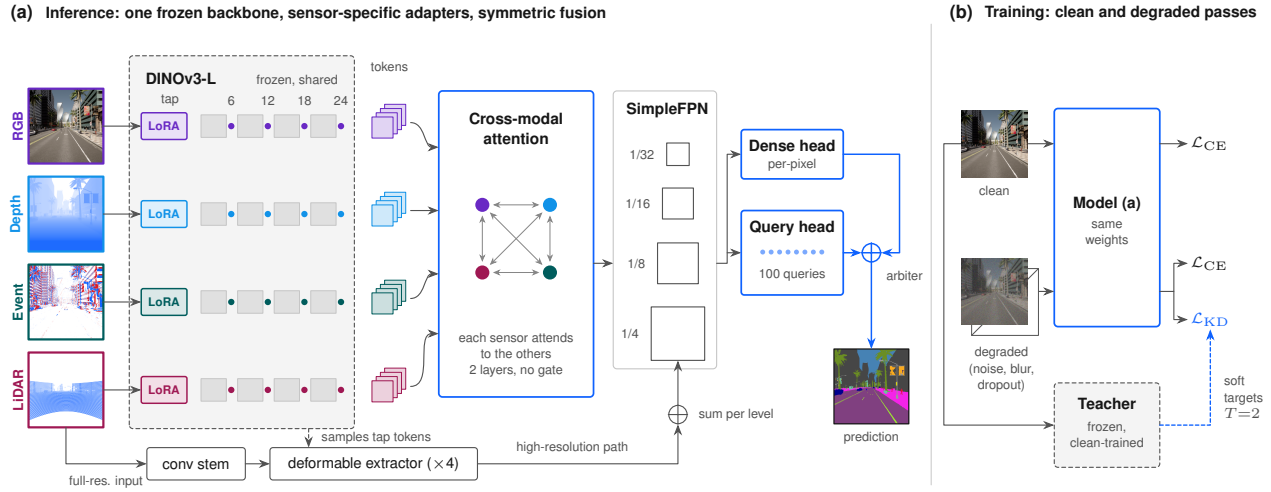


FIGURE 1. Overview of ReliaDINO trained with C2D2. (a) Each sensor passes through the same frozen DINOv3-L with its own LoRA adapter; tokens from four depths are fused by cross-modal attention, decoded by a SimpleFPN augmented with a high-resolution path, and classified by a dense head and a query head combined by an arbiter. (b) C2D2: every batch is used twice, a clean pass and a synthetically degraded pass, the latter also distilled from a frozen teacher trained on clean inputs. Blue outlines mark trained components; dashed boxes are frozen. [TODO: add the sensor router to the drawing or caption (FU-01); redraw per notes/figure_guide.md]

operation. The fused map is the average of the attended maps plus a sensor-specific residual,

$$F = \frac{1}{S}\sum_s \hat{T}_s + \sum_s \tanh(\gamma_s)\, m_s(T_s), \tag{2}$$

where m_s is a small MLP and γ_s is a learned scalar per sensor, initialized at 0.1.

C. SENSOR ROUTER

Each sensor also has an auxiliary head that predicts class logits A_s from T_s and is trained with cross-entropy. A per-class router mixes these predictions pixel by pixel. Its weights are

$$w_{s,k}(p) = \mathrm{softmax}_s\left(h_{s,k}(T_s)(p) + \lambda\, r_s(p)\right), \tag{3}$$

where $r_s = 1 - H(\sigma(A_s))/\log K$ is the normalized confidence of the auxiliary prediction of sensor s (with gradients stopped), K is the number of classes, and h_s is a small network whose last layer is initialized at zero, so that routing starts from confidence alone. The routed logits $R_k = \sum_s w_{s,k} A_{s,k}$ are added to the final prediction with a learned scale α initialized at zero (Section IV-E). The router is supervised with cross-entropy on R and with a regularizer that favours decisive per-pixel weights. [TODO: role of the router after FU-01 (DRN-261029-06/07/08)]

D. DECODER WITH A HIGH-RESOLUTION PATH

A SimpleFPN [20] turns F into a feature pyramid at strides 4, 8, 16, and 32. Because every level is interpolated from stride-16 tokens, structures thinner than a token are poorly represented. We therefore add a high-resolution path that follows the adapter of the official DINOv3 segmentation code [13], [20]. A convolutional stem shared across sensors, followed by sensor-specific 1×1 projections, extracts multi-scale features from the raw inputs. Deformable-attention extractors, one for each tapped block and two more after the last, let these convolutional features sample semantic evidence from the tapped token maps. Unlike the original adapter, nothing is injected back into the backbone, which stays untouched. The output of the path is added to the FPN levels.

E. DENSE AND QUERY HEADS

A dense convolutional head classifies every pixel at stride 4 and gives logits D. A Mask2Former-style query head [22] with 100 queries and six decoder layers attends to the token maps of all sensors, tagged with a sensor embedding, and its class and mask predictions are converted into per-pixel class scores Q. The final logits are

$$\ell_k(p) = D_k(p) + \mathrm{softplus}(\lambda_k)\, Q_k(p) + \alpha\, R_k(p), \tag{4}$$

with a learned per-class weight λ_k. During training, each step uses the dense head alone, the query head alone, or both, with probabilities 0.25, 0.25, and 0.5, so that neither head can rely on the other; at inference both are used.

F. CLEAN-TO-DEGRADED TWO-PASS DISTILLATION

C2D2 uses every training batch (x, y) twice.

Clean pass. The model sees the clean inputs x and is trained with the segmentation loss $\mathcal{L}_{\mathrm{clean}}$, which combines online hard-example cross-entropy [25] on ℓ with the losses of the query head, the auxiliary sensor heads, and the router, and two regularizers listed in Section V.

Degraded pass. The model sees a degraded copy $\tilde{x}$. Each sensor of each sample is corrupted independently with probability 0.5. One fifth of the corrupted sensors are removed entirely by setting their normalized input to zero; the others receive one operator drawn from a sensor-specific set: patch dropout, blur, and gain and gamma changes for all image-like sensors, a colour shift for RGB, holes for depth, dropped

beams for LiDAR, and resolution loss for events. The corruption strength is capped at 0.3, 0.6, and 1.0 of its maximum from epochs 0, 10, and 20 onward. Gaussian and salt-and-pepper noise are never used in training, so that the noise evaluation tests corruption that the model has not seen. The degraded pass is supervised by the labels and by a frozen teacher g, a model of the same architecture without the high-resolution path that was trained on clean inputs with a longer schedule. The teacher sees the clean input:

$$\mathcal{L}_{\text{deg}} = \mathcal{L}_{\text{ce}}\big(\ell(\tilde{x}), y\big) + \beta\, T^2\, \mathrm{KL}\Big(\sigma\Big(\tfrac{g(x)}{T}\Big) \,\Big\|\, \sigma\Big(\tfrac{\ell(\tilde{x})}{T}\Big)\Big), \quad (5)$$

with $\beta = 0.5$ and $T = 2$, where σ is the softmax over classes and the Kullback–Leibler divergence is averaged over pixels. The total loss is $\mathcal{L}_{\text{clean}} + \mathcal{L}_{\text{deg}}$.

C2D2 needs no condition labels and no degradation labels: the degradation is sampled on the fly, and the teacher only sees clean inputs. It changes training only; the model at inference is the same network run once. [TODO: after FU-03 (L7, L5, L6): one paragraph on what each part contributes and how C2D2 differs from distillation with missing modalities [8], [9]]

V. EXPERIMENTS

A. IMPLEMENTATION DETAILS

All sensors are converted to three-channel images as in CMNeXt [1]. We train for 40 epochs with AdamW (learning rate 6×10^{-4}, weight decay 0.01), a polynomial schedule with power 0.9, and a 10-epoch warm-up, on 768×768 crops with an effective batch size of 16 obtained by gradient accumulation. We use the data augmentation of DGFusion [6] so that both models see the same augmented images. The clean-pass loss combines online hard-example cross-entropy [25] on the final logits with the set-prediction loss of the query head (weight 0.5, with deep supervision), the auxiliary sensor heads (0.5), the router (cross-entropy 0.4 and regularizer 0.01), a class-prototype contrastive term (0.1), and a variance-covariance regularizer on the per-sensor tokens (weight 1.0 for LiDAR and 0.25 for the other sensors). These terms are inherited from earlier versions of our model; we list them for reproducibility and report them in the supplementary material. The frozen teacher is the same architecture without the high-resolution path, trained for 200 epochs on clean data. Unless noted, we report the mean of three seeds and the best seed; ablations use one seed.

B. COMPARISON WITH PRIOR FUSION MODELS

Table 2 compares ReliaDINO with fusion models on DELIVER. We list only methods evaluated under the protocol that reports test-split mIoU, because DELIVER results in the literature are reported under at least three different protocols and are not comparable across them. ReliaDINO reaches 57.74 test mIoU on average over three seeds and 58.61 for the best seed, 1.0 and 1.9 points above the published DGFusion result; against the official DGFusion weights evaluated with our code (56.73), the mean gap is 1.01. Its margin over MM SAM-adapter, which uses only RGB and depth, is 0.4 points on average On the validation set, ReliaDINO scores 70.17 on average, 1.4 points above CAFuser-CAA.. Tables 3 and 4 report the same model on the real MUSES and MCubeS benchmarks [TODO: results pending: MUSES four-sensor model, MCubeS transfer].

TABLE 2. DELIVER (RGB, depth, event, LiDAR), mIoU (%).

Method	Backbone	Sensors	val	test
CMNeXt todo	todo	RGB-D-E-L	todo	todo
StitchFusion todo	todo	RGB-D-E-L	todo	todo
GeminiFusion todo	todo	RGB-D-E-L	todo	todo
CAFuser todo	todo	RGB-D-E-L	todo	todo
CAFuser-CAA todo	todo	RGB-D-E-L	68.79	todo
DGFusion todo	Swin-T	RGB-D-E-L	todo	56.71
Fewer sensors (secondary comparison)				
MM SAM-adapter todo	todo	RGB-D	69.60	57.35
Ours (3-seed mean)	DINOv3-L	RGB-D-E-L	70.17	57.74
Ours (best seed)	DINOv3-L	RGB-D-E-L	69.59	58.61

todo: original table/page per baseline; protocol (val-best checkpoint, 1024, nearest-exact label resampling); DINOv3-L is frozen.

TABLE 3. MUSES, official evaluation, mIoU (%).

Method	Backbone	Sensors	val	test
CAFuser todo	todo	todo	todo	todo
DGFusion todo	todo	todo	todo	79.5
Camera only or fewer sensors (secondary comparison)				
GtA todo	todo	RGB	todo	82.39
MM SAM-adapter todo	todo	RGB-L	todo	81.07
Ours	DINOv3-L	RGB-E-L-R	todo	todo

C. ROBUSTNESS TO MISSING AND DEGRADED SENSORS

Table 5 and Fig. 2 evaluate all models under the same missing-sensor and noise protocol. Under the public benchmark definition on the validation set, ReliaDINO scores 57.85 EMM, 12.9 points above MAGIC, the strongest model reported by the benchmark under the same zero-fill definition [11], and it keeps 61 to 66 mIoU when a quarter to three quarters of a sensor's pixels are missing. On the test set, the official DGFusion weights fall from 56.73 to 31.14 EMM and to 4.5 mIoU under the strongest noise, and our reproduction of DGFusion behaves the same way (31.53 and 5.52). Retraining DGFusion and CAFuser with our synthetic degradation raises their EMM to 45.5 and 46.5 without changing their architecture. ReliaDINO reaches 48.7 EMM while keeping the highest clean accuracy, and it retains 47.9 mIoU under the strongest noise, which was never seen during training.

When only one sensor is available (Fig. 2b), the official DGFusion weights collapse to 7.5 mIoU with RGB alone and to 1.1 with event or LiDAR alone, while still reaching 43.3 with depth alone. Training with degraded inputs lifts every single-sensor score, most strongly for RGB (49.0 for the retrained DGFusion and 53.9 for our model without the high-

TABLE 4. MCubeS (RGB, AoLP, DoLP, NIR), test mIoU (%).

Method	Backbone	test
Mul-VMamba todo	todo	54.65
StitchFusion todo	todo	55.9
Ours	DINOv3-L	todo

TABLE 5. Robustness to missing and degraded sensors on DELIVER, mIoU (%).

(a) Public missing-modality benchmark, val

Method	EMM	RMM .25	RMM .50	RMM .75
MAGIC [7][§]	44.97	51.61	48.19	45.30
MAGIC++ [15][§]	44.85	53.92	49.31	47.06
Ours (3 seeds)	57.85	65.53	63.35	61.28

(b) Same protocol for all models, test

Method	Clean	EMM	S&P .05	S&P .10	S&P .20
DGFusion (official)*	56.73	31.14	39.12	16.85	4.53
DGFusion (reprod.)[‡]	55.68	31.53	37.97	16.71	5.52
DGFusion (retrained[†])	54.77	45.52	48.89	40.51	23.34
CAFuser (retrained[†])	55.79	46.50	todo	todo	todo
Ours w/o C2D2	todo	40.01	todo	todo	todo
Ours w/o high-res	56.59	48.06	53.25	51.64	48.79
Ours	57.74	48.66	53.11	51.36	47.85

[§] Values reported by the robustness benchmark [11] (MiT-B2 backbone, same zero-fill definition). Methods that skip a missing input instead of zero-filling it report results under a different protocol and are not listed. * Official released weights evaluated with our code; their S&P values are the clean score minus the measured drop. [‡] Our reproduction with the official recipe, without degradation. [†] Retrained with the same synthetic degradation. Ours: mean of 3 seeds; baselines: single runs. EMM, RMM, and S&P are defined in Section III.

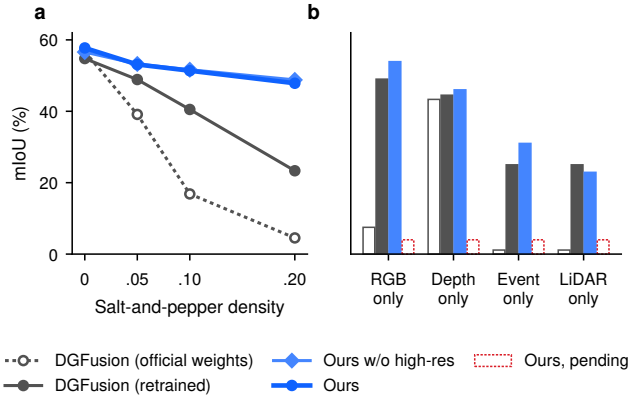


FIGURE 2. Robustness on DELIVER test under one protocol. (a) Salt-and-pepper noise applied to every sensor. (b) Only one sensor available (the other three zero-filled). Ours: mean of three seeds; DGFusion: single runs, official weights or retrained with the same synthetic degradation. [TODO: (b) full-model values pending.]

TABLE 6. DELIVER test mIoU (%) per condition, all sensors present.

Model	Sun	Cloud	Rain	Fog	Night	Low exp.	All
DGFusion (official)	todo	todo	todo	todo	todo	todo	56.73
DGFusion (retrained[†])	todo	todo	todo	todo	todo	todo	54.77
CAFuser (retrained[†])	todo	todo	todo	todo	todo	todo	55.79
Ours w/o high-res	todo	todo	todo	todo	todo	todo	56.59
Ours	todo	todo	todo	todo	todo	todo	57.74

[†] Retrained with synthetic degradation. All: same values as Table 5(b) clean column.

resolution path).

D. QUALITATIVE RESULTS

Figs. 3 and 4 show scenes selected by a fixed rule: among all test images, those where the gap between our model and the best baseline is largest, at most one scene per condition and map [TODO: describe what the scenes show once selected].

E. ACCURACY PER CONDITION

Table 6 breaks clean test accuracy down by condition [TODO: text after the per-condition numbers arrive (analysis request 20261028-183826 item 1)].

F. ABLATION

Table 7 builds the model up from a frozen DINOv3-L with LoRA on RGB alone and then removes one component at a time from the full model, all with the same seed and schedule. The frozen backbone alone reaches 55.14 mIoU with RGB, already within 1.6 points of DGFusion with four sensors, so a large part of the accuracy comes from the foundation model. Adding the other three sensors with cross-modal attention and the query head gives 55.94, the two-pass training with distillation adds 1.8 points (57.73), and the high-resolution path adds 0.9 (58.61). Removing cross-modal attention or the query head costs 0.85 and 0.72 points, respectively. Replacing the sensor-specific adapters with one shared adapter does not reduce clean accuracy (58.84) [TODO: robustness of the shared variant and the degradation-pass and distillation rows (L5, L6) pending].

G. COMPUTATIONAL COST

Table 8 reports parameters, operations, latency, and memory at the evaluation resolution [TODO: numbers pending (DRN-261027-02); the two-pass training doubles the forward passes during training but adds nothing at inference].

VI. ANALYSIS

A. WHERE SENSOR OVER-RELIANCE COMES FROM

A fusion model can lean on one sensor because its architecture or supervision favours that sensor, or because the training data make that sensor sufficient. DGFusion [6] is a useful test case: it adds depth supervision to its fusion, and removing depth costs our reproduction of DGFusion 23 to 37 points depending on the condition (Fig. 5). If the depth supervision caused this dependence, a model without it should behave differently.

Our own model trained without C2D2, which has no depth supervision, shows the same pattern: removing depth costs 23.5 to 38.7 points, and the cost grows from sun to low exposure as for DGFusion. The same architecture trained with C2D2 loses only 3.9 points in sun, 10.1 at night, and 13.4 under low exposure. The effect also appears in a second architecture: when we re-train DGFusion with the same syn-

todo: Fig. 3 when a sensor fails: rows = depth missing / RGB missing / LiDAR+event missing / RGB with heavy noise; columns = affected input, ground truth, DGFusion (official), DGFusion (retrained), CAFuser (retrained), ours; error pixels in one colour

FIGURE 3. [TODO: caption; state the scene-selection rule (top differences between ours and the best baseline, one scene per condition, no repeated map)]

todo: Fig. 4 adverse conditions, all sensors present: DELIVER night / fog / rain / low exposure and MUSES real night / snow / fog; columns = RGB, ground truth, DGFusion, CAFuser, ours; zoomed crops where the baselines miss objects

FIGURE 4. [TODO: caption; state the scene-selection rule]

TABLE 7. Ablation on DELIVER test (one seed, 40 epochs), mIoU (%).

Configuration	Clean	EMM
Building up		
Frozen DINOv3-L + LoRA, RGB only	55.14	–
+ depth, event, LiDAR (mean fusion)	todo	todo
+ cross-modal attention, query head	55.94	todo
+ C2D2 (two-pass training, distillation)	57.73	48.68
+ high-res path (full model)	58.61	49.12
Removing one component from the full model		
− sensor-specific LoRA (one shared)	58.84	todo
− cross-modal attention (mean fusion)*	57.76	todo
− query head and arbiter	57.89	todo
− degradation pass	todo	todo
− teacher distillation	todo	todo

*Also removes the trunk expansion. todo: EMM column for ablation runs; decision on shared LoRA pending its robustness test.

TABLE 8. Computational cost on DELIVER at 1024 × 1024.

Model	Params (M)	Trainable (M)	GFLOPs	Latency (ms)	Memory (GB)
DGFusion	todo	todo	todo	todo	todo
CAFuser	todo	todo	todo	todo	todo
Ours w/o high-res	todo	todo	todo	todo	todo
Ours	todo	todo	todo	todo	todo

thetic degradation, removing depth entirely costs 4.7 points, whereas it costs 33.3 points for our reproduction trained without degradation and 33.4 for the official weights (Table 5).

These measurements suggest that, on DELIVER, the dependence on depth comes mainly from training without degraded sensors rather than from a particular fusion design or supervision [TODO: final wording after FU-04 (C2D2 applied to DGFusion and CAFuser) and FU-05 (all curves on one split)]. A plausible reason is that DELIVER renders depth without noise or holes, so training on complete inputs finds a shortcut that real depth sensors would not offer. The comparison has two caveats: the model trained without C2D2 was trained for 200 epochs and the C2D2 model for 40, and the DGFusion curve was measured on the test split while ours were measured on validation. A complementary observation is that our model trained without C2D2 is already less sensitive to salt-and-pepper noise than the re-trained DGFusion; we therefore attribute part of the noise robustness to the frozen DINOv3 backbone rather than to C2D2 [TODO: state the measured drops (ledger:117: E1 3.09/5.73/11.22 vs DGFusion retrained 5.9/14.3/31.4) after checking that the definitions match].

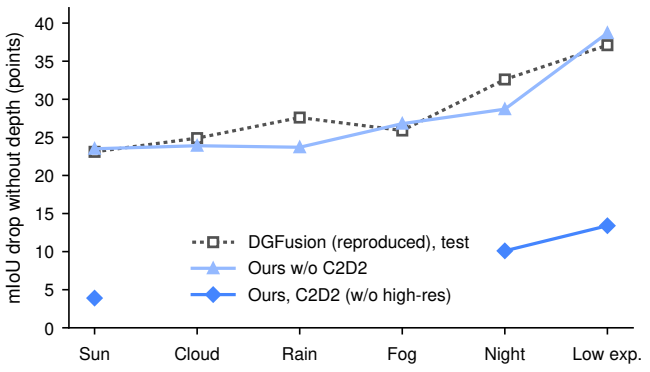


FIGURE 5. **Drop in mIoU when depth is removed, per condition. Our model trained without C2D2 relies on depth as heavily as our reproduction of DGFusion, whose design adds depth supervision; the same architecture trained with C2D2 loses much less. DGFusion was measured on test, the other curves on val. [TODO: re-measure all curves on one split; add the full model and the retrained baselines.]**

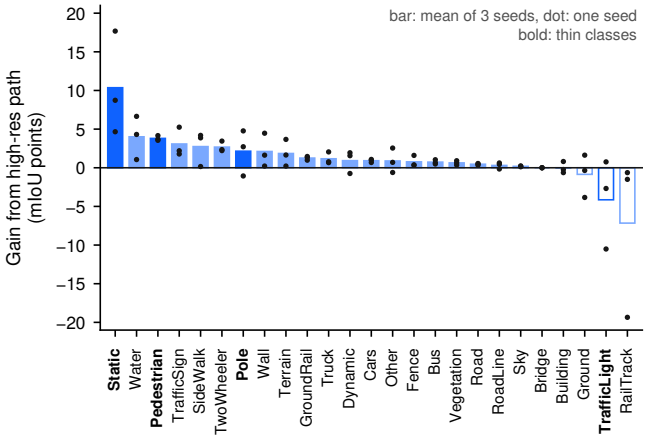


FIGURE 6. **Per-class gain of the high-resolution path on DELIVER test, paired by seed (full model minus the same model without the path). Thin classes gain most; TrafficLight and RailTrack lose and vary strongly across seeds. [TODO: add zoomed crops of thin objects (Q2 dumps on yeon).]**

B. WHAT THE HIGH-RESOLUTION PATH ADDS

The high-resolution path raises clean test mIoU by 1.15 points on average over three seeds (from 56.59 to 57.74; Table 7). Fig. 6 shows where the gain comes from. Averaged over the three seeds, Static gains 10.4 points, Pedestrian 3.8, TrafficSign 3.1, and Pole 2.1, and the four thin classes gain 3.0 points together. The gain comes mainly from recall: for Pole the recall rises from 0.75 to 0.78 and for Pedestrian from 0.89 to 0.90 [TODO: confirm these recall values (two seeds, NAS analysis_logs/a2_thin_anatomy_20261018/summary.md) through the analysis session before citing]. Two classes lose: TrafficLight by 4.1 points and RailTrack by 7.2. RailTrack varies strongly between seeds even without the path (58.5, 32.6, and 45.1 for the three seeds), so we do not interpret its change. The TrafficLight loss, in contrast, is systematic and is analysed next.

todo: limitation: posters and signs predicted as traffic light (2 of the 4 recorded scenes; RGB, ground truth, ours, error map)

FIGURE 7. **[TODO: caption; selection rule from the scene README: among the 20 test images where our model adds the most TrafficLight false positives relative to the model without the high-resolution path, one scene per condition.]**

C. FAILURE CASE: POSTERS READ AS TRAFFIC LIGHTS

With the high-resolution path, the recall of TrafficLight rises (0.80 to 0.85) but its precision falls (0.46 to 0.41) [TODO: same confirmation as above]. Fig. 7 shows where the extra false positives appear. In the scenes with the largest increase, billboards and posters, labelled as *Other*, are painted as traffic lights, and in one rainy scene a traffic sign is. The two classes share fine structure, a small bright rectangular shape on a pole or a wall, but differ in context. Because the high-resolution features are added to every pyramid level without a gate, fine detail reaches the dense per-pixel classifier directly, and the classifier can follow the detail without enough context. This mechanism is a hypothesis that our measurements are consistent with but do not prove; we discuss a remedy in Section VII.

D. LIMITATIONS

First, DELIVER is synthetic, and its complete depth is the reason the shortcut exists; the size of the effect on real depth sensors remains to be measured, and MUSES does not contain depth. Second, all our runs use a 40-epoch schedule and three seeds, while the baselines are single runs, so differences below about one point should not be over-read. Third, the high-resolution path makes the model slightly more sensitive to heavy pixel noise and to partial loss of depth or RGB than the same model without it (Table 5); a likely cause is that its convolutional stem sees raw pixels, which we have not yet verified. Fourth, a single shared LoRA reached the same clean accuracy as sensor-specific adapters in one seed (Table 7); [TODO: update once the shared-LoRA robustness test decides which variant is the paper model].

VII. CONCLUSION

We proposed C2D2, a two-pass training scheme that adds a synthetically degraded pass, distilled from a frozen clean-input teacher, to ordinary training. Applied to a fusion model on a frozen DINOv3 backbone with sensor-specific adapters, cross-modal attention, and a high-resolution path, it gives the most accurate four-sensor result on DELIVER that we know of while keeping the model robust when sensors are removed

or corrupted [TODO: and on MUSES and MCubeS, once measured (FU-08)]. Our analysis indicates that the heavy dependence of fusion models on depth in DELIVER comes mainly from training without degraded sensors [TODO: wording after FU-04 and FU-05]. It also exposes what remains: the high-resolution path recovers thin objects but lets fine detail mislead the per-pixel classifier, as in posters read as traffic lights, which suggests routing fine detail to mask prediction rather than to per-pixel classification. Measuring the same dependence on real depth sensors is the next step that our synthetic-data findings call for.

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[TODO: AUTHOR NAME] [TODO: biography (photo 1 in × 1.25 in)]